Lovepreet Sharma

AI/ML Engineer | Software Developer | M.Tech Data Analytics, NIT Jalandhar

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Summary

Software engineer with production experience across the full stack and a focus on applied AI systems. Have shipped end-to-end AI products: a multi-agent plant-disease diagnosis system that cuts harmful automated decisions from 76% to 14% by routing low-confidence cases to humans, a document RAG platform (FastAPI + FAISS + Next.js), and federated deep-learning experiments on 38-class PlantVillage. Equally comfortable building conventional software — delivered a fully offline loan management system now used daily by a finance business. Strong CS fundamentals (GATE CS/IT qualified 2024, 2025, 2026); awarded a Pre-Placement Offer during internship.

Education

Dr. B R Ambedkar National Institute of Technology Jalandhar — Punjab, India
M.Tech in Data Analytics — *2025 – Present*

Giani Zail Singh Campus College of Engineering & Technology — Bathinda, Punjab
B.Tech in Computer Science and Engineering — *CGPA: 7.85* — *2021 – 2025*

Experience

ThinkNEXT Technologies Pvt. Ltd. — Jan 2025 – Jun 2025
Machine Learning Engineer Intern — *Mohali, Punjab*

- Built AI-powered product features in Python end to end — data preprocessing, feature engineering, model training, and evaluation — using Scikit-learn, Pandas, and NumPy.
- Designed modular backend services and REST APIs that served trained models to production applications, keeping inference, data access, and business logic cleanly separated.
- Ran exploratory data analysis, feature selection, and error analysis to improve predictive performance, and presented findings through visual reports to non-technical stakeholders.
- Worked with SQL databases and backend services powering data-driven applications and ML pipelines; collaborated via Git with code review, testing, and documentation.
- Awarded a **Pre-Placement Offer (PPO)** for technical performance, project ownership, and problem-solving ability.

Infowiz Pvt. Ltd. — Jun 2023 – Jul 2023
Software Development Intern — *Mohali, Punjab*

- Developed backend modules for a Hospital Management System in PHP and MySQL with secure role-based authentication and a normalized relational schema.
- Implemented patient registration, appointment scheduling, billing, doctor management, and admin operations, then optimized SQL queries to cut page-load times on record-heavy screens.
- Partnered with senior developers on debugging, testing, and deployment in an Agile workflow.

Projects

Agentic AI for Plant Disease Diagnosis | *Python, PyTorch, Multi-Agent, FAISS, LLM* — **GitHub**

- Designed a multi-agent system that wraps a ResNet18 PlantVillage classifier (38 classes) with a vision agent, a confidence/quality agent, and an orchestrator implementing a three-tier routing policy: direct diagnosis, retrieve-and-reason, or escalate to a human expert.
- Reduced the **harmful-action rate from 75.8% (naive always-trust-the-model baseline) to 14.2%** over a 190-image evaluation by abstaining and escalating when confidence, prediction margin, or ensemble agreement was low.
- Grounded every automated diagnosis in a retrieval layer (bge-small embeddings + FAISS over an agronomy corpus) so LLM responses cite scientific sources instead of hallucinating treatment advice.
- Added image-quality gating (Laplacian blur and brightness checks) and a 3-model ensemble whose disagreement signal caught confident misclassifications that thresholding alone missed; shipped an automated evaluation harness reporting accuracy, macro-F1, correct-action rate, and intervention rate.

RAG Engine — Document Q&A Platform | *FastAPI, LangChain, FAISS, Next.js, Docker* — **GitHub**

- Built a production-style Retrieval-Augmented Generation platform that ingests PDF, DOCX, HTML, Markdown, CSV, and JSON documents and answers questions with page-level citations; one-command deploy via Docker Compose.

– Implemented hybrid retrieval combining dense FAISS search with BM25 through reciprocal rank fusion, plus optional cross-encoder reranking (bge-reranker-base), improving answer relevance over pure vector search.
– Engineered the full query pipeline — normalization, metadata-filter extraction, query-type classification, retrieval confidence scoring, and post-generation answer validation — with streaming (SSE) conversational responses and session memory.
– Made the LLM layer provider-agnostic (OpenAI, Anthropic, OpenRouter, Ollama) behind a factory interface, and added JWT auth with roles, rate limiting, security headers, structured logging, per-provider cost tracking, and a background job queue.

Federated Learning for Plant Disease Detection | *PyTorch, FedProx/FedAvg, ResNet50* **GitHub**

– Implemented a 5-client FedProx simulation on the PlantVillage dataset with ResNet50 — 50 communication rounds, 10 local epochs per round, mixed-precision training — logging per-round accuracy, precision, recall, and F1.
– Benchmarked federated optimizers (FedAvg, FedProx, FedOpt, FedMA, SCAFFOLD) under non-IID client splits to quantify the accuracy and convergence cost of preserving data privacy.
– Authored a review paper on privacy-preserving federated learning for precision agriculture, analyzing communication efficiency, client heterogeneity, and convergence behaviour across published agricultural datasets.

Loan Management System | *Node.js, SQLite, Express, JavaScript, NSIS* *Commercial product*

– Built and shipped a fully offline, zero-install loan management desktop application handling customers, loan schemes, disbursement, repayment ledgers, penalties, income/expense accounting, and printable receipts, statements, and agreements.
– Implemented four interest models (reducing-balance EMI, flat, fixed-installment, interest-free) across daily, weekly, fortnightly, monthly, and custom-interval schedules, with automatic payment allocation across penalty, interest, and principal.
– Guaranteed financial correctness by recomputing balances from immutable payment history rather than mutating them in place, backed by a 35-assertion automated test suite over the money math and ledger.
– Packaged the app as a portable Windows bundle and NSIS installer with an embedded Node runtime and a GitHub Actions build pipeline, plus one-click backup/restore with a staged, crash-safe restore path.

Technical Skills

Languages: Python, Java, C++, C, JavaScript, SQL

AI & Machine Learning: Machine Learning, Deep Learning, NLP, Large Language Models, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Federated Learning, Computer Vision, Prompt Engineering, Model Evaluation

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, FastAPI, Express.js, Pandas, NumPy

Data & Storage: MySQL,PostgreSQL, FAISS,

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, System Design

Tools & Practices: Git, GitHub, GitHub Actions (CI/CD), Docker, Linux, REST API design, Unit Testing, Agile, Android Studio, Jupyter Notebook

Achievements

- Qualified **GATE CS/IT** in three consecutive years (2024, 2025, 2026).
- Received a **Pre-Placement Offer (PPO)** during Machine Learning Engineering internship at ThinkNEXT Technologies Pvt. Ltd.
- Authored a review research paper, **Federated Learning for Plant Disease Detection**, surveying state-of-the-art privacy-preserving learning algorithms.
- Delivered a production loan management system currently in daily use by a small finance business.
- Awarded a merit scholarship for academic excellence in Senior Secondary Education.